

A-MES

AUTOMOTIVE MES • SEOYON E-HWA

V O L 0 0

Project Overview & Data Foundation

Index • ERD • DB Schema

Document • L3 Detailed Design Specification

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Reference • Plant A " SAV Detroit, MI | Plant B " GEO Birmingham, AL

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TABLE OF CONTENTS

Project Overview & Data Foundation

INDEX	A-MES UI Design Specification — Index
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AMES_ERD	A-MES ERD
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AMES_DB_Schema	A-MES Database Schema
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● AUTOMOTIVE MES · UI DESIGN SPECIFICATION

A-MES

MANUFACTURING EXECUTION SYSTEM

자동차 부품 제조실행시스템 UI 설계서

VERSION

3.1

CLIENT

서연이화

PLANT

SAV / GEO

VOLUMES

VOL.03 – VOL.13

11

MODULES / 모듈

2

BOP ROUTINGS

3

QC GATES

10

REPORT TYPES

29

MASTER DATA
ENTITIES

7

RBAC ROLES

150+

BUSINESS RULES

OVERALL PROCESS FLOW 전체 공정 흐름

SCM
(ERP)WH
창고PP
생산계획INJ/IMG/PNT
생산공정QC
품질FG
완제품

출하

ROUTING A

도어트림 / 콘솔어퍼



ROUTING B

내장 도색 사출품 (TBD)



VOL03

WH

Warehouse Mgmt

창고관리

Receiving / PDA / Inventory / Location

입고 · PDA · 재고 · 로케이션 관리

VOL04

PP

Production Plan

생산계획

Work Order / MRP / Line Schedule / OEE

작업지시 · MRP · 라인스케줄 · 가동/OEE

VOL05

INJ

Injection

사출생산

POP / Defect / Mold Shot Count

POP · 불량 · 금형 쇼트수 관리

VOL06

IMG

Wrapping

감싸기생산

Fabric Roll / PDA 4-Step / Defect

원단롤 · PDA 4단계 · 불량

VOL07

PNT

Painting

VOL08

QC

Quality Control

도장생산

5-Step Process / Oven Monitor / Defect

5단계공정 · 건조로모니터 · 불량

품질관리

3-Gate Inspection / Enhanced Flag

3게이트검사 · 강화검사 플래그

VOL09

FG

Finished Goods

완제품출고

FIFO Picking / Delivery / Shipping

FIFO피킹 · 거래명세서 · 출고

VOL10

MNT

Maintenance

설비관리

OEE / PM Calendar / Mold Block

OEE · PM달력 · 금형 강제차단

VOL11

RPT

Reports

분석리포트

10 Reports / Report Center / Builder

10종리포트 · 리포트센터 · 빌더

VOL12

SYS

System Admin

시스템관리

RBAC 7-Role / Config / Health / Audit

7역할RBAC · 환경설정 · 헬스 · 감사로그

VOL13

MD

Master Data

기준정보

29 Entities / 5 Categories / Table Design

29개 엔티티 · 5카테고리 · 테이블설계

SITE MAP

사이트맵

—

전체

페이지

구조

VOL03 · WH

Warehouse

창고관리

8 screens · L2 →

WH-01

Inbound Schedule

WH-02

PDA Inbound

WH-03

Inventory Status

WH-04

Location Map

WH-05

Inventory Adjust

WH-06

Release Schedule

WH-07

PDA Release

WH-08

Transaction History

VOL04 · PP

Production Plan

생산계획

13 screens · L2 →

PP-01

Forecast Mgmt

PP-02

SAP B1 Import

PP-03

Plan & Confirm

PP-04

Work Order Mgmt

PP-05

MRP

PP-06

Purchase Req.

PP-07

WO Release

PP-CAL

Production Calendar

PP-0TD

On-Time Delivery

PP-LSB

Line Schedule Board

PP-0DM

Downtime Monitor

PP-0EE

Line Availability & OEE

PP-DTL

Downtime Log Entry

VOL05 · INJ

Injection

사출생산

8 screens · L2 →

INJ-01

POP Login

INJ-02

POP Dashboard

INJ-03

WO Confirmation

INJ-04

Production Entry

INJ-05

Defect Register

INJ-06

Mold Change

INJ-07

Production Status

INJ-08

Andon Emergency

VOL06 · IMG

Wrapping

감싸기생산

7 screens · L2 →

IMG-01

POP Login

IMG-02

POP Dashboard

IMG-03

Production Entry

IMG-04

Fabric Input & Verify

IMG-05

Defect Register

IMG-06

Bond Setup

IMG-07

Production Status

VOL07 · PNT

Painting

도장생산

9 screens · L2 →

PNT-01

Daily Production Plan

PNT-02

Virtual LOT Pre-Issue

PNT-03

Loading Station (R1)

PNT-04

Line Tracking Board

PNT-05

Oven Monitor (R2)

PNT-06

Unloading & Count (R3)

PNT-07

LOT Label Apply

PNT-08

Defect Register

PNT-09

Shift Log / Report

VOL08 · QC

Quality Control

품질관리

9 screens · L2 →

QC-01

Incoming Inspection

QC-02

In-Process Inspection

QC-03

Final Inspection

QC-04

NCR Register

QC-05

Hold / Quarantine

QC-06

QC-07

QC-08

QC-TRC

CAPA Management

QC Dashboard

Inspection Standard

Traceability

V0L09 · FG

Finished Goods

완제품출고

10 screens · L2 →

FG-01

PDA Stocking

FG-02

FG Inventory

FG-03

Shipment Order

FG-04

PDA FIFO Picking

FG-05

Loading Confirmation

FG-06

Delivery Note

FG-07

Day-End Close

FG-08

Shipment History

FG-09

FG Dashboard

FG-RTN

Customer Return

V0L10 · MNT

Maintenance

설비관리

9 screens · L2 →

MNT-01

Equipment Card

MNT-02

Failure Register

MNT-03

OEE Analysis

MNT-04

Mold Management

MNT-05

PM Schedule

MNT-06

Downtime Log

MNT-07

Work Order

MNT-08

Spare Parts

MNT-09

MNT Dashboard

V0L11 · RPT

Reports

분석리포트

10 screens · L2 →

RPT-01

Daily Production

RPT-02

Defect Pareto

RPT-03

Daily Shipment

RPT-04

On-Time Delivery

RPT-05

Inventory Status

RPT-06

Equipment OEE

RPT-07

Monthly KPI

RPT-08

Schedule Adherence

RPT-09

Report Center

RPT-10

Report Builder

V0L12 · SYS

System Admin

시스템관리

8 screens · L2 →

SYS-01

User Management

SYS-02

RBAC Permissions

SYS-03

Factory Calendar

SYS-04

Interface Monitor

SYS-05

Audit Log

SYS-06

Notification Settings

SYS-07

System Configuration

SYS-08

System Health

V0L13 · MD

Master Data

기준정보 · 29 엔티티

5 hubs + 29 entities · L2 →

CATEGORY L3 HUBS · 카테고리 허브

L3 · A

Foundation

L3 · B

Routing & Production

L3 · C

Resources & Equipment

L3 · D

RFID & Materials

L3 · E

Quality & Logistics

ENTITY TABLE-DESIGN PAGES · 엔티티 테이블 설계서

MD-01

Item Master

MD-02

BOM

MD-03

BOM Version

MD-04

BOP Routing

MD-05

Work Center

MD-06 Inspection Std	MD-07 Vendor	MD-08 Equipment	MD-09 Mold	MD-10 Paint / Fabric
MD-11 Shipment Dest	MD-12 Customer	MD-13 UOM	MD-14 Calendar	MD-15 Jig
MD-16 RFID Tag	MD-17 RAL Paint Color	MD-18 Oven	MD-19 RFID Reader	MD-20 Line Master
MD-21 Defect Code	MD-22 Defect Cause	MD-23 Packaging Spec	MD-24 Label Template	MD-25 Reason Code
MD-26 Common Code Group	MD-27 Spare Parts	MD-28 PM Template	MD-29 Line Time Pattern	

REF

Data Model

데이터 모델 참조

2 references

ERD
Entity Relationship Diagram

DB
Database Schema

BUSINESS RULE OVERVIEW

핵심 업무 규칙

MODULE	RULE	규칙 요약	ACTION
WH	BR-WH-001	입고 수량 $\pm 5\%$ 초과 시 SCM 승인 요청	Block / Alert
WH	BR-WH-002	로케이션 용량 초과 시 입고 차단	Block
PP	BR-PP-001	부품 재고 부족 시 WO 차단 (MRP)	Block / Alert
PP	BR-PP-002	WO 확정 후 자재 자동 예약	Auto-Reserve
INJ	BR-INJ-001	불량률 $\geq 5\%$ 시 라인 자동 정지	Stop / Alert
INJ	BR-INJ-002	금형 쇼트수 90% $\rightarrow$ 경고, 100% $\rightarrow$ 강제차단	Warn / Block
IMG	BR-IMG-001	원단 색상 불일치 시 PDA 다음 단계 차단	Block
PNT	BR-PNT-001	5단계 공정 순서 강제 적용 (건너뛸 불가)	Block
PNT	BR-PNT-002	건조로 온도 이탈 시 Andon 발동	Alert / Andon
QC	BR-QC-001	3회 연속 불량 시 강화검사 플래그 자동 설정	Flag
QC	BR-QC-002	QC FAIL $\rightarrow$ 해당 WO 완료 차단	Block

FG	BR-FG-001	FIFO 원칙: 입고일 오래된 로트 우선 출고	Sort / Alert
MNT	BR-MNT-001	OEE < 65% 시 설비 점검 알림	Alert
SYS	BR-SYS-001	로그인 5회 실패 시 계정 잠금	Lock
MD	BR-MD-001	Phase-0 미완료 품목은 WO 생성 차단	Block
MD	BR-MD-33	라인 시간패턴 세그먼트는 0~1440분을 빈틈·중첩 없이 분할	Block
PP	BR-PP-031	WO 블록은 단일 OPERATING 가동창 안에 완전히 배치	Block
PP	BR-PP-035	계획 가동창 내 설비 정지는 실제 비가동(UNPLANNED) 자동 분류	Auto
PP	BR-PP-042	OEE = 가용성 × 성능 × 품질, 기간별 스냅샷 저장	Auto

MODULE INTEGRATION 모듈 연계

FROM	TO	DATA / EVENT	설명
SCM (ERP)	WH	PO Sync	발주 정보 자동 수신
SCM (ERP)	PP	Customer Order	수주 수신 → 생산계획 트리거
WH	PP	Stock Level	재고 수준 실시간 공유
PP	INJ / IMG / PNT	Work Order	WO 전달 → POP 화면 표시
INJ / IMG / PNT	QC	Production Result	생산 완료 → QC 검사 트리거
QC	FG	PASS Flag	합격 제품 → 완제품 창고 이동
FG	SCM (ERP)	Shipment Confirm	출하 완료 → ERP 전송
INJ	MNT	Mold Shot Count	금형 쇼트수 → 설비관리 모니터링
All	RPT	Transaction Data	전 모듈 데이터 → 리포트 집계
MD	ALL	Master Data	기준정보 → 전 모듈에 배포
MD-29	PP	Line Time Pattern	라인별 분단위 시간패턴 → 스케줄·가동 기준

DATA MODEL REFERENCES 데이터 모델 참조

ERD

ERD

Entity Relationship Diagram

전체 엔티티 관계도

Cross-module table relationships

전 모듈 테이블 관계 다이어그램

DB

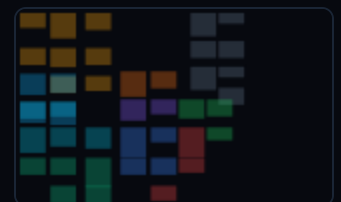
SCHEMA

Database Schema

데이터베이스 스키마

Full table / column DDL definitions

전체 테이블·컬럼 DDL 정의



A-MES Database Schema

MSSQL · .NET · 자동차 부품 MES · 간략 설계도

PK FK 클릭: 모듈 필터

ALL MD 기준정보 WH 창고 PP 생산계획 INJ 사출 IMG 감싸기 PNT 도장 QC 품질
FG 완제품 MNT 설비 SYS 시스템

MD – MASTER DATA 기준정보

MD MD_Customer 거래선	MD MD_Item 품목
PK CustomerID INT IDENTITY CustomerCode VARCHAR(20) SAV / GEO CustomerName NVARCHAR(100) Plant VARCHAR(10) Savannah/G... IsActive BIT DEFAULT 1	PK ItemID INT IDENTITY ItemCode VARCHAR(30) UNIQUE ItemName NVARCHAR(200) ItemType VARCHAR(20) FG/WIP/MAT... Unit VARCHAR(10) EA / KG / M FK CustomerID INT → MD_Cust... RoutingType CHAR(1) A / B Phase0Complete BIT DEFAULT 0
MD MD_BOM 자재명세	MD MD_BOP 공정순서
PK BOMID INT IDENTITY FK ParentItemID INT → MD_Item FK ChildItemID INT → MD_Item Qty DECIMAL(10,4) EffectiveDate DATE	PK BOPID INT IDENTITY FK ItemID INT → MD_Item StepSeq TINYINT 1,2,3... ProcessCode VARCHAR(10) INJ/IMG/PN... FK WorkCenterID INT → MD_Work... StdTime DECIMAL(8,2) 분
MD MD_WorkCenter 설비/작업장	MD MD_Location 로케이션
PK WorkCenterID INT IDENTITY WCCode VARCHAR(20) UNIQUE WCName NVARCHAR(100) ProcessCode VARCHAR(10) Capacity INT 일생산능력	PK LocationID INT IDENTITY LocationCode VARCHAR(20) A-01-01 Zone VARCHAR(10) WH / FG / ... MaxQty INT 최대재재 IsActive BIT

WH – WAREHOUSE 창고관리

WH WH_PurchaseOrder 발주(SCM)	WH WH_Receiving 입고
PK POID INT IDENTITY PONo VARCHAR(30) SCM P0번호 FK CustomerID INT → MD_Cust... PODate DATE Status VARCHAR(20) Pending/Re...	PK ReceivingID INT IDENTITY FK POID INT → WH_Purc... FK ItemID INT → MD_Item LotNo VARCHAR(30) ReceivedQty DECIMAL(10,2) FK LocationID INT → MD_Loca...

		ReceivedAt	DATETIME
		FK UserID	INT → SYS_User
WH	WH_Inventory	재고 (현재고)	
PK	InventoryID	INT	IDENTITY
FK	ItemID	INT	→ MD_Item
FK	LocationID	INT	→ MD_Loca...
	LotNo	VARCHAR(30)	
	OnHandQty	DECIMAL(10,2)	
	ReservedQty	DECIMAL(10,2)	WO 예약량
	UpdatedAt	DATETIME	
● PP – PRODUCTION PLAN 생산계획			
PP	PP_CustomerOrder	수주	
PK	OrderID	INT	IDENTITY
	OrderNo	VARCHAR(30)	SCM 수주번호
FK	CustomerID	INT	→ MD_Cust...
FK	ItemID	INT	→ MD_Item
	OrderQty	INT	
	DueDate	DATE	
	Status	VARCHAR(20)	Open/Closed
PP	PP_WorkOrder	작업지시	
PK	WOID	INT	IDENTITY
	WONo	VARCHAR(30)	UNIQUE
FK	OrderID	INT	→ PP_Cust...
FK	ItemID	INT	→ MD_Item
	PlanQty	INT	
	PlanStartDate	DATE	
	Status	VARCHAR(20)	Draft/Rele...
	IsBlocked	BIT	MRP 차단
● INJ / IMG / PNT – PRODUCTION 생산실적 (공통구조)			
PROD	PR_ProductionResult	생산실적	
PK	ResultID	INT	IDENTITY
FK	WOID	INT	→ PP_Work...
	ProcessCode	VARCHAR(10)	INJ / IMG ...
FK	WorkCenterID	INT	→ MD_Work...
	GoodQty	INT	양품
	DefectQty	INT	불량
	StartTime	DATETIME	
	EndTime	DATETIME	
FK	UserID	INT	→ SYS_User
PROD	PR_DefectDetail	불량 상세	
PK	DefectID	INT	IDENTITY
FK	ResultID	INT	→ PR_Prod...
	DefectCode	VARCHAR(20)	INJ-D01 등
	DefectQty	INT	
	Note	NVARCHAR(500)	
PNT	PNT_ProcessStep	도장 단계 이력	
PK	StepLogID	INT	IDENTITY
FK	ResultID	INT	→ PR_Prod...
	StepSeq	TINYINT	1~5
	StepName	VARCHAR(30)	탈지/하도/...
	CompletedAt	DATETIME	
	OvenTemp	DECIMAL(5,1)	건조로 온도...
● QC – QUALITY CONTROL 품질관리			

QC_QC_Inspection검사 헤더			
PK	InspectionID	INT IDENTITY	
FK	ResultID	INT → PR_Prod...	
FK	WOID	INT → PP_Work...	
	Gate	TINYINT 1 / 2 / 3	
	InspectedQty	INT	
	PassQty	INT	
	FailQty	INT	
	Result	VARCHAR(10) PASS / FAIL	
	EnhancedFlag	BIT 강화검사 여부	
FK	InspectorID	INT → SYS_User	
	InspectedAt	DATETIME	
FG – FINISHED GOODS 완제품출고			
FG_FG_Stock완제품 재고		FG_FG_Shipment출하	
PK	FGStockID	INT IDENTITY	
FK	ItemID	INT → MD_Item	
FK	LocationID	INT → MD_Loca...	
	LotNo	VARCHAR(30)	
	OnHandQty	INT	
	InboundDate	DATE FIFO 기준	
PK	ShipmentID	INT IDENTITY	
FK	OrderID	INT → PP_Cust...	
FK	FGStockID	INT → FG_Stock	
	ShipQty	INT	
	ShipDate	DATETIME	
	DeliveryNote	VARCHAR(30) 거래명세서 ...	
FK	UserID	INT → SYS_User	
MNT – MAINTENANCE 설비관리			
MNT_MNT_Equipment설비 기준		MNT_MNT_PMSchedulePM 계획	
PK	EquipmentID	INT IDENTITY	
FK	WorkCenterID	INT → MD_Work...	
	EqCode	VARCHAR(20)	
	MoldShotLimit	INT 금형 한계 ...	
	CurrentShotCount	INT 누적 쇼트수	
	Status	VARCHAR(20) Active/PM/...	
PK	PMScheduleID	INT IDENTITY	
FK	EquipmentID	INT → MNT_Equ...	
	PlannedDate	DATE	
	PMTYPE	VARCHAR(20) Daily/Week...	
	Status	VARCHAR(20) Plan/Done/...	
MNT_MNT_OEELogOEE 실적			
PK	OEELogID	INT IDENTITY	
FK	EquipmentID	INT → MNT_Equ...	
	LogDate	DATE	
	Availability	DECIMAL(5,2) %	
	Performance	DECIMAL(5,2) %	
	Quality	DECIMAL(5,2) %	
	OEE	AS (A*P*Q) Computed	
SYS – SYSTEM 시스템 / 공통			
SYS_SYS_User사용자		SYS_SYS_Role역할 (RBAC)	

PK	UserID	INT IDENTITY	
	LoginID	VARCHAR(50)	UNIQUE
	PasswordHash	VARCHAR(256)	BCrypt
FK	RoleID	INT	→ SYS_Role
	FailCount	TINYINT	5회 잠금
	IsLocked	BIT	
	IsActive	BIT	

SYS

SYS_AuditLog

감사로그

PK	LogID	BIGINT IDENTITY	
FK	UserID	INT	→ SYS_User
	Action	VARCHAR(50)	INSERT/UPD...
	TableName	VARCHAR(50)	
	RecordID	INT	
	OldValue	NVARCHAR(MAX)	JSON
	NewValue	NVARCHAR(MAX)	JSON
	CreatedAt	DATETIME	5년 보관

총 24개 테이블 MD: 6 · WH: 3 · PP: 2 · PROD: 3 · QC: 1 · FG: 2 · MNT: 3 · SYS: 3 · PNT: 1

MSSQL 2019+ · .NET 8 EF Core